SMART HOSPITAL AVAILABILITY & EMERGENCY ROUTING SYSTEM

AI-Powered Healthcare Decision Support System

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Critical Challenges in Emergency Healthcare

- In emergency situations, patients rush to the nearest hospital without proper information
- However, many hospitals may lack:
  - Available ICU beds
  - Specialized doctors (e.g., cardiologist, neurologist)
  - Critical medical equipment
- This results in:
  - Delayed treatment
  - Increased medical complications
  - Higher risk to patient life

"Right patient at the wrong hospital can cost a life."

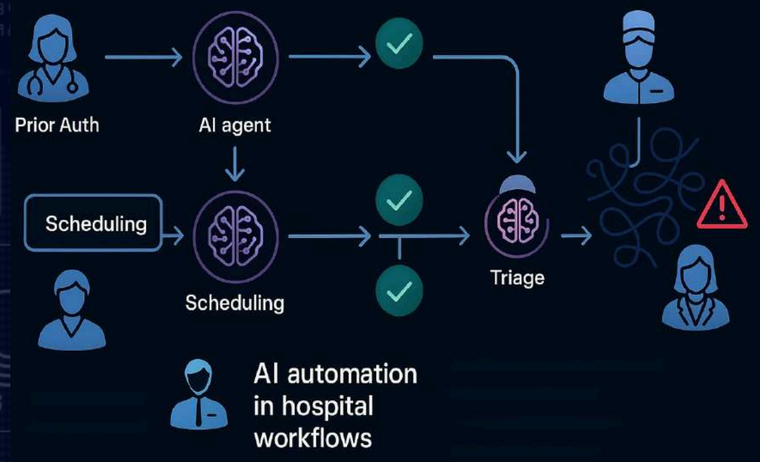
Gap in Current Emergency Healthcare System

Current System Limitations:

- No real-time information about hospital availability
- No system to check ICU beds, doctors, or equipment
- Patients rely only on distance, not suitability
- No communication between patient and hospital before arrival

There is no intelligent system that connects patient needs with real-time hospital capabilities.

Proposed Smart Solution



- We propose an AI-powered emergency healthcare system that:
  - Analyzes patient symptoms in real-time
  - Determines the severity level (Critical / Moderate / Normal)
  - Identifies hospitals with required facilities (ICU, specialists, equipment)
  - Recommends the most suitable hospital, not just the nearest one
  - Sends patient data to the hospital before arrival

"Delivering the right care at the right place, at the right time."

Methodology

- User inputs symptoms through the application
- The system performs AI-based analysis to determine severity
- It checks a database for real-time hospital availability
- Matches patient needs with required facilities (ICU, specialists, equipment)
- Recommends the most suitable hospital
- Sends patient details to the hospital before arrival

AI-Based Decision System

- The system uses an AI-based decision model to analyze patient symptoms
- It classifies cases into three categories:
- Critical – Immediate attention required (ICU / specialists)
- Moderate – Urgent but manageable
- Normal – Basic medical care needed
- Based on severity, the system recommends:
- Best-equipped hospital (for critical cases)
- Nearest suitable hospital (for moderate cases)
- General hospitals (for normal cases)

"AI ensures faster, smarter, and more accurate decision-making in emergencies."

Intelligent Hospital Matching System

- The system matches patient requirements with hospital capabilities
- It considers:
  - Availability of ICU beds
  - Presence of specialized doctors
  - Required medical equipment
- Ensures that the patient is directed to the most suitable hospital
- Reduces chances of wrong hospital selection and treatment delays

Example:

Input: Chest pain + high BP

System Output:

Hospital A (Cardiologist + ICU available)

Hospital B (No specialist availab

Pre-Arrival Alert & Smart Communication

- The system sends patient details to the selected hospital before arrival
- Shared information includes:
  - Patient symptoms and condition
  - Severity level (Critical / Moderate / Normal)
- The hospital can:
  - Accept and prepare for the patient in advance
  - Redirect if required resources are unavailable
- Enables faster response and immediate treatment upon arrival

System Demonstration Flow

Example Scenario:

- User Input: “Chest pain, sweating, high BP”
- AI Analysis: High-risk (Critical condition detected)
  - System Recommendation:
  - Hospital A (Cardiologist + ICU available)
  - Hospital B (No specialist available)
- Pre-Arrival Alert: Patient data sent to Hospital A
- Outcome: Hospital is prepared before patient arrival

Technology Stack

- Frontend: React.js / HTML, CSS
- Backend: Node.js / Python
- Database: Firebase / JSON-based storage
- AI Module: Rule-based logic / Machine Learning model
- APIs: Google Maps API (for location & routing)

Result and Impact

- Reduces emergency response time
- Minimizes wrong hospital visits
- Ensures faster and more accurate treatment
- Improves patient survival chances
- Enhances coordination between patient and hospital
- Can reduce treatment delays by up to 30–50% (estimated)

Future Scope

- Integration with real-time hospital availability systems
- Ambulance tracking and smart dispatch system
- Advanced map-based navigation with live updates
- Integration with government emergency services (e.g., 108)
- Support for voice-based input for faster access

Conclusion

- Our system provides a smart and efficient solution for emergency healthcare
- It uses AI to make faster and more accurate decisions
- Helps patients reach the most suitable hospital in time
- Reduces delays and improves overall emergency response

"By combining AI with healthcare, we aim to save time, improve decisions, and ultimately save lives."

THANK YOU